

JENIS PATEL

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Research Profile

I am an ecologist with research interests in spatial ecology, species distribution modelling, and conservation biology. My work focuses on understanding how ecological processes operating at multiple spatial and temporal scales shape biodiversity patterns, and how quantitative models can inform conservation decisions. I have experience with hierarchical Bayesian models, biogeographic modelling, multi-species occupancy modelling, movement ecology, landscape-scale habitat suitability mapping using satellite imagery, and GPS telemetry. I work primarily in R and Google Earth Engine, with growing proficiency in Python for geospatial data processing. My research spans vertebrate ecology (carnivores, birds) and freshwater invertebrate biomonitoring.

Education

PhD in Carnivore Ecology

2019–2024

Nature Conservation Foundation, Bangalore & Manipal Academy of Higher Education, Manipal

Thesis: A multi-scale study of carnivore ecology in the high mountains of Asia. Used biogeographic models (DEC, BAMM), multi-species occupancy models, and GPS telemetry-based movement models to study carnivore community assembly across regional, landscape, and local scales.

MSc Environmental Science

2012–2014

Veer Narmad South Gujarat University, Surat

Dissertation: Odonata assemblages and water quality in urban lakes. Identified pollution-tolerant indicator species for rapid freshwater assessments.

BSc Biotechnology

2007–2011

Veer Narmad South Gujarat University, Surat

Professional Experience

Project Consultant, Endangered Birds Programme

2024–2025

Nature Conservation Foundation (NCF), Bangalore

Developed landscape-scale habitat suitability models for Jerdon's courser (~300,000 km²) using logistic regression, canonical correlation analysis, and 64-band Google Satellite Embeddings. Built an interactive Google Earth Engine application for conservation planning. Responsibilities also included data analysis, report writing, manuscript preparation, and transferring quantitative skills to team members through documented workflows and regular teaching sessions.

Independent Researcher, Forest Owlet Project

2015–2016

Dangs, Gujarat (funded by Rufford and BNHS/IBCN)

Secured independent funding. Conducted baseline distribution mapping and threat assessments for the critically endangered Forest Owlet. Engaged with local tribal farming communities to integrate local ecological knowledge into survey design. Conducted hierarchical Bayesian occupancy modelling for the Forest Owlet across three nested spatial scales.

Research Biologist

2014–2015

Voluntary Nature Conservancy (VNC)

Assessed Indian grey wolf occupancy using sign surveys and local interviews. Conducted socio-economic surveys on human–wolf interactions and developed mitigation strategies.

Wildlife Biologist

2014

Wild India, Ranchi, Jharkhand

Collaborated with the Jharkhand State Forest Department on tiger and co-predator monitoring. Trained forest department personnel in camera trapping, sign surveys, GIS mapping, and prey density estimation.

Scientific Publications

Lead-authored

Patel, J. et al. (2026) Landscape-scale habitat suitability mapping for the critically endangered Jerdon's courser (*Rhinoptilus bitorquatus*). *Submitted to Remote Sensing in Ecology and Conservation*. Conceived study, developed models, built GEE application, wrote manuscript.

Patel, J. et al. (2026) Scale-dependent habitat selection reveals conditional compatibility between Forest Owlet conservation and traditional agroforestry in Gujarat, India. Pre-print on BioRxiv; to be submitted to *Biological Conservation*. Designed study, collected field data, fitted hierarchical Bayesian occupancy models, wrote manuscript.

Patel, J. et al. (2024) Snow leopards or solar parks? Reconciling wildlife conservation and green energy development in the high Himalaya. *Biological Conservation*. Led analysis of spatial overlap between conservation areas and energy infrastructure, wrote manuscript.

Patel, J. et al. (2023) Influence of predator suppression on prey availability on carnivore occurrence in western Himalaya. *Journal of Zoology*. Designed study, analysed camera trap data using multi-species occupancy models, wrote manuscript.

Patel, J. et al. (2015) New distribution record of the forest owlet *Heteroglaux blewitti* in Purna Wildlife Sanctuary, Gujarat, India. *JOTT*. Fieldwork, species identification, wrote manuscript.

Co-authored

Are Galliformes of the high Himalayas well protected? Identifying conservation priority areas using an assemblage-level approach. (2026) *Integrative Conservation*.

Estimating snow leopard and prey populations at large spatial scales. (2021) *Ecological Solutions & Evidence*.

Population assessment of the Endangered Nilgiri tahr using the double-observer survey method. (2021) *Oryx*.

Diversity of Odonata in protected areas of Gujarat, India. (2022) *Journal of Biological Control*.

New records of polymorphism in Asian libellulid dragonflies. (2020) *International Journal of Odonatology*.

Popular Articles

Active by Day, Asleep by Night: The Forest Owlet of Dangs. (2022) *NavGujarat Samay*. (in Gujarati)

Defenders of the mound. (2017) *The Hindu*.

Pedagogical Qualifications and Outreach

Carnivore Ecology Course

2024

Designed and led a five-day training programme on population ecology, community ecology, and conservation strategies for 20+ students from across India.

Forest Owlet Monitoring Workshop

2023

Organised training for 35 forest officials (Range Forest Officers, Beat Guards, forest watchers) on Forest Owlet ecology, distribution, and threats. Co-organised with Voluntary Nature Conservancy and North Dang Forest Division, Gujarat.

Forest Owlet Conservation Action Plan

2020

Contributed talks and brainstorming sessions as part of the MOEFCC action plan developed by SACON.

Skills transfer and mentoring

2024–2025

Regular teaching sessions on statistical modelling in R and geospatial analysis for team members at NCF. Documented and archived reproducible analytical workflows.

Grants and Awards

Rufford Small Grant for Nature Conservation (€5,753)

2023

Principal Investigator.

BNHS/Indian Bird Conservation Network Grant (€714)

2015–2016

Principal Investigator. Assessment of status, distribution and conservation of Forest Owlet in Dangs, Gujarat.

Analytical and Technical Skills

Programming: R (statistical modelling, data visualisation, spatial analysis), Google Earth Engine (land cover classification, change detection, satellite embedding analysis), QGIS (vector processing), Python (geospatial data handling, pre-trained models for large-scale analysis).

Statistical modelling: Hierarchical Bayesian occupancy models, multi-species occupancy models, logistic regression, canonical correlation analysis, biogeographic models (DEC, BAMM), GPS movement models (step-selection, resource selection functions), species distribution models.

Field methods: Camera trapping (occupancy, SECR, N-mixture), radio/GPS telemetry, passive acoustic monitoring (BirdNET, Raven Pro, Perch 2.0), sign surveys, vegetation surveys.

References

Kulbhushansingh Suryawanshi, Ph.D. (PhD supervisor)

Director, India program of the Snow Leopard Trust

Scientist, Nature Conservation Foundation, CIFAR fellow

Email: kulbhushan@ncf-india.org

Abishek Harihar, PhD (PhD collaborator)

Director, Tiger Program, Panthera

Email: aharihar@panthera.org